

Theo Marsh

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Education

Cedarholt Polytechnic Institute, Ashford Bend May 2026
Bachelor of Science in Computer Engineering; Minor in Applied Statistics GPA: 3.6/4.0
Coursework: Software Testing and Verification, Computer Systems, Data Structures and Algorithms,
Robotics System Design

Projects

Visionserver, Ridgeline Robotics Nov 2021 – Jan 2024
C++, C, NASL

- Built a C++ vision-processing library for Raspberry Pi deployment on a competition robot, spanning nearly 6,000 files across C, C++, and NASL.
- Configured continuous integration to validate builds, sustaining active development across 200 commits over 86 days.

UE5 Lidarsim, Independent Project Oct 2023 – May 2024
C++, C#, C

- Built a lidar simulation in Unreal Engine 5 to model obstacle detection for a competition robot, in C++ and C#.
- Iterated over 38 active days and 55 commits, coordinating changes across a multi-language codebase of roughly 300 files.

Swerve Simulation, Ridgeline Robotics Sep 2023 – Jan 2024
Java

- Extended a Java swerve-drive simulation from an earlier robotics project, adding drivetrain calculations validated with JUnit and the Jama matrix library.
- Wrote JUnit test suites covering the simulation logic, sustaining development across 51 files over 33 active days.

Activities & Extracurriculars

Ridgeline Robotics, Vision Systems Contributor Sep 2021 – Jan 2024

- After teammates struggled to reproduce intermittent vision failures during practice runs, began logging the conditions behind each failure before proposing a fix.
- Traced late-season failures to code that had never been exercised outside competition, shifting focus toward testing before deployment.

Awards & Accomplishments

Regional Robotics Tournament — Technical Inspection Award Mar 2023

Skills & Interests

Language: C/C++, Java, JavaScript

Frontend: HTML/CSS

DevOps: CI/CD

Quality: Unit Testing

Product: Technical Writing